DESIGN MILESTONE

EMPATHIZE

I investigated hot beverage preparation as a daily ritual through literature review, interviews, and observations. The research explored how current machines reduce this experience to automated efficiency.

DEFINE

The main tension identified was between automation efficiency and the need for a psychological ritual pause in daily routines.

IDEATE

Different concept directions were generated to explore how ritual engagement and subtle automation could coexist. These ideas focused on structuring the brewing process as a meaningful micro-ritual while maintaining efficiency and usability.

PROTOTYPE

Physical and digital prototypes were developed to explore the interaction, system structure, and ritual sequence of the brewing process. These prototypes helped test how the concept could translate into a guided ritual experience.

TESTING

Both the physical and digital prototypes were tested to evaluate the interaction flow and ritual experience. Feedback helped refine how the system supports psychological transition while maintaining subtle automation.

MILESTONE 1: EMPATHIZE

Ritual culture | Beverage traditions | Social interaction | Cultural meaning

Selected Impactful Prompt
I am currently researching ritual-based hot beverage preparation practices. Could you recommend academic articles, books, or reliable sources about the history and cultural significance of ritual-based hot beverages such as coffee or tea?

AI Role
Expander

AI Response Summary
AI suggested historical and cultural examples of beverage rituals such as Ethiopian coffee ceremonies, Japanese tea ceremonies, and ancient cacao rituals.

Reflection
This research helped me understand that hot beverage preparation historically carries cultural and social meaning beyond pure functionality.

✓ Accepted
Ritual preparation as a conceptual foundation for the project.

✕ Rejected
Viewing beverage machines purely as functional appliances.

🔄 Transformed
Coffee machine → ritual-oriented beverage preparation system.

Selected Impactful Prompt
"Coffee's Origin Stories & Spread — I analysed this article from ResearchGate and would like to learn more from this. Give me more articles like this."

AI Role
Expander

AI Response Summary
AI recommended additional academic articles discussing the origin, cultural spread, and historical meaning of coffee across different societies.

Reflection
The additional academic sources strengthened my understanding of the cultural and historical background of coffee rituals.

✓ Accepted
Using academic and historical research to support the conceptual direction.

✕ Rejected
Limiting the research only to contemporary product analysis.

🔄 Transformed
Product-focused research → cultural and ritual-focused research.

MILESTONE 2: DEFINE

Design tension | Psychological transition | Ritual vs automation | Design direction

Selected Impactful Prompt

I am currently researching ritual-based hot beverage preparation practices. Could you recommend academic articles, books, or reliable sources about the history and cultural significance of ritual-based hot beverages such as coffee or tea?

AI Role
Clarifier

AI Response Summary

AI categorized the tensions emerging from the research according to frequency and psychological impact. It identified key tensions such as automation efficiency vs ritual engagement and blurred daily transitions vs the need for a psychological boundary.

Reflection

This helped me clearly see which tensions were most meaningful and recurring across my research.

✓ Accepted
The tension between automation efficiency and meaningful ritual engagement.

✕ Rejected
Focusing only on technical or maintenance-related problems as the main design driver.

🔄 Transformed
Hot beverage preparation → a psychological transition ritual within daily routines.

Selected Impactful Prompt

"After looking at my research which direction decision I can lead to? Give multiple choices for me to analyze."

AI Role
Expander

Reflection

Exploring multiple directions helped me evaluate how technology could support the ritual without dominating the experience.

AI Response Summary

AI proposed several potential design directions such as ritual-structured interaction, slow interaction systems, and automation-driven solutions.

✓ Accepted
A ritual-focused system where technology subtly supports the preparation process.

✕ Rejected
Fully automated systems that remove user involvement from the ritual.

🔄 Transformed
Automation → subtle technological support within a slow-living ritual experience.

MILESTONE 5: IDEATE

Concept generation | Ritual interaction | System-level design | Automation vs ritual

Selected Impactful Prompt

I Generate at least three concept variations for a ritual-focused hot beverage device. Each concept should clearly respond to the selected design tension and explore different system-level approaches rather than purely formal variations. The concepts should align with the framing logic developed in my research.

AI Role
Expander

AI Response Summary

AI generated multiple concept directions that respond to the selected design tension and explore different ways of structuring the brewing process as a ritual experience.

Reflection

This helped me explore different conceptual approaches and understand how the brewing process could be structured as a meaningful ritual rather than only a functional activity.

✓ Accepted

Concept directions that structure the brewing process as a ritual experience.

✕ Rejected

Concepts focused only on formal variations without a clear experiential or conceptual logic.

🔄 Transformed

Coffee machine → ritual-oriented brewing system

Selected Impactful Prompt

Concept 1 is not expressing the ritual experience as strongly as I expected. Could you help me reframe this concept so that it emphasizes a micro-ritual experience while still maintaining efficiency in the preparation process?

AI Role
Re-framer

AI Response Summary

AI reinterpreted Concept 1 and transformed it into Condensed Micro-Ritual Engine, structuring the brewing process as a short but intentional ritual sequence while maintaining efficiency.

Reflection

Through this iteration I realized that the original concept version focused too much on symbolic mode switching rather than an embodied ritual sequence. Reframing the concept helped clarify how efficiency and ritual pacing can coexist through a compressed micro-ritual structure.

✓ Accepted

Condensed Micro-Ritual Engine as the refined version of Concept 1.

✕ Rejected

The original Boundary Mode concept because it treated the transition mainly as a functional mode change.

🔄 Transformed

Concept 1 → Condensed Micro-Ritual Engine emphasizing a micro-ritual sequence.

MILESTONE 4: PROTOTYPE

Concept generation | Ritual interaction | System-level design | Automation vs ritual

Selected Impactful Prompt

Based on this physical prototype, what improvements could be made to the physical prototype to make the brewing workflow clearer, more intuitive, and more ergonomic? Please focus on the placement of the grinder, filter, water tank, and cup area, and suggest changes that would improve the interaction between grinding, brewing, and serving while maintaining a minimal design.

AI Role
Critique

AI Response Summary

AI suggested improving the spatial hierarchy of the components. The grinder should be positioned slightly higher to clarify the grinding-to-brewing flow. A guiding channel between the grinder and filter could make the coffee transfer cleaner and more visible. The water tank connection to the brewing area could be visually clarified to communicate the brewing process. Additionally, an adjustable filter height could allow both cup and carafe usage, and a clear interaction point such as a single brew button could improve usability.

Reflection

AI helped me evaluate the prototype from a workflow and interaction perspective. It highlighted how component placement influences the clarity of the brewing sequence and helped me identify improvements that would strengthen both usability and the ritual experience.

✓ Accepted

A clearer grind → brew → serve workflow that strengthens the ritual sequence while keeping the system visually understandable.

✗ Rejected

Raising the grinder position too high, because it would move the system away from the pour-over brewing logic and weaken the ritual structure of the preparation process.

🔄 Transformed

Prototype feedback → refining the spatial relationship between grinder, filter, and cup area to support a micro-ritual brewing sequence while maintaining a minimal design.

MILESTONE 5:TESTING

User testing | Ritual experience | Interaction feedback | System clarity

Selected Impactful Prompt

In user testing, participants interpreted blue as preparation, green as ready, and red as warning. How could the light feedback system be refined to support a calm ritual experience without creating stress? Please explain with different options so I can analyze them.

AI Role
Critique / Expander

AI Response Summary

AI suggested that the existing color system already communicates the brewing stages clearly, but the warning signal could be refined to avoid disturbing the ritual atmosphere. Possible refinements included adjusting the brightness of the red light, using a softer pulsing behavior, or limiting the warning signal to specific situations.

Reflection

This analysis helped me evaluate how light behavior affects the emotional perception of the brewing process. Since my concept focuses on a calm and ritual-oriented experience, the warning signal should communicate attention without creating stress.

✓ Accepted

Clear color-based communication of brewing stages.

✕ Rejected

Strong or constantly illuminated warning signals that disturb the ritual experience.

🔄 Transformed

Red warning light behavior refined to appear in a more subtle and controlled way.

Selected Impactful Prompt

I selected the upper right carafe for my coffee machine because its cylindrical glass body provides stability and visual clarity during brewing. The combination of glass, wood, and a dark base also aligns with the minimal and ritual-oriented aesthetic of my system.
Please evaluate this carafe within the context of a ritual-focused pour-over coffee set. Please analyze its strengths and weaknesses in terms of form language, stability, pouring ergonomics, and visual coherence with the machine. Then suggest possible improvements while maintaining a minimal and refined design.

AI Role
Critique

AI Response Summary

AI evaluated the carafe as a suitable choice because of its clear form language and material harmony with the system. It suggested refining the pouring spout for cleaner liquid flow, improving handle ergonomics for better pouring control, and slightly adjusting body proportions to enhance both stability and visual elegance.

Reflection

This evaluation helped me understand how small adjustments in spout design, handle geometry, and proportions can improve usability while maintaining the minimal aesthetic of the set.

✓ Accepted

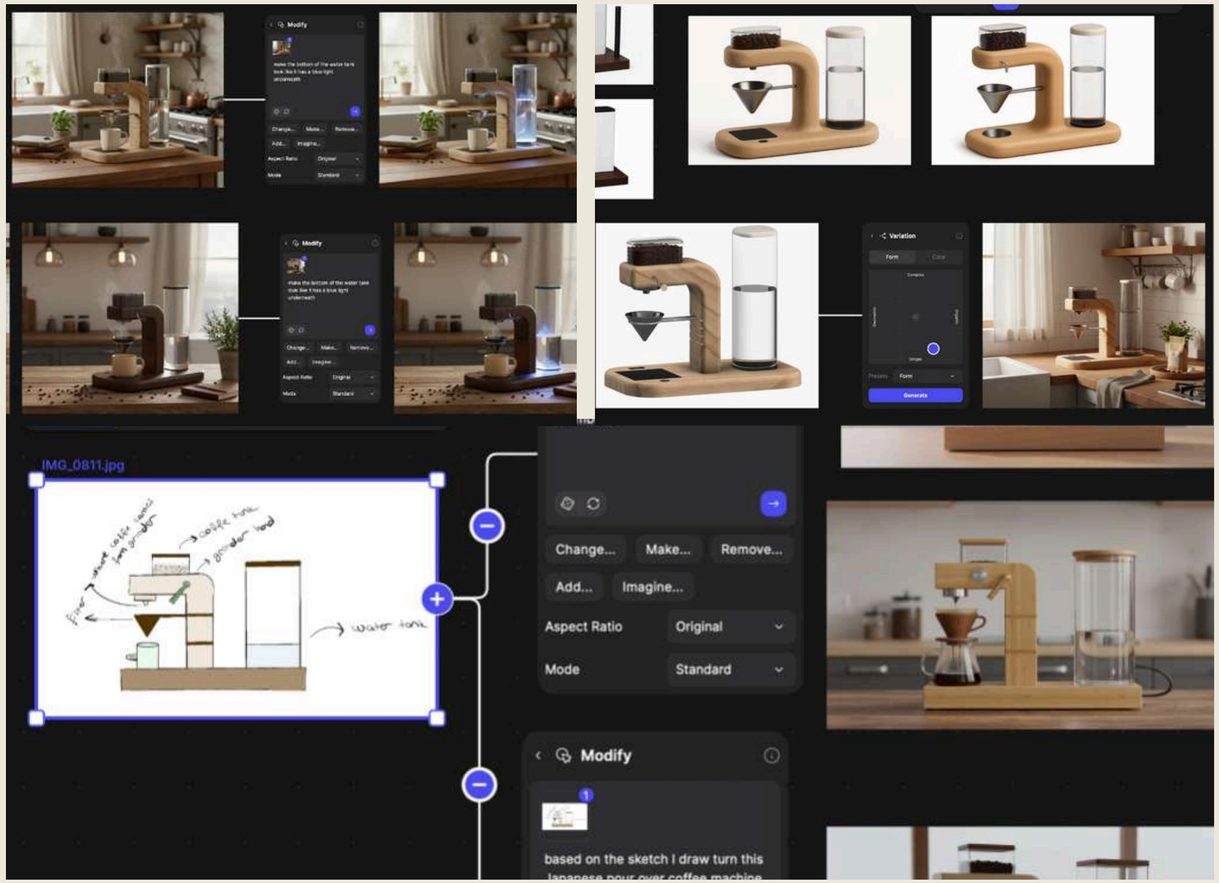
Transparent cylindrical body and material combination that supports the ritual aesthetic.

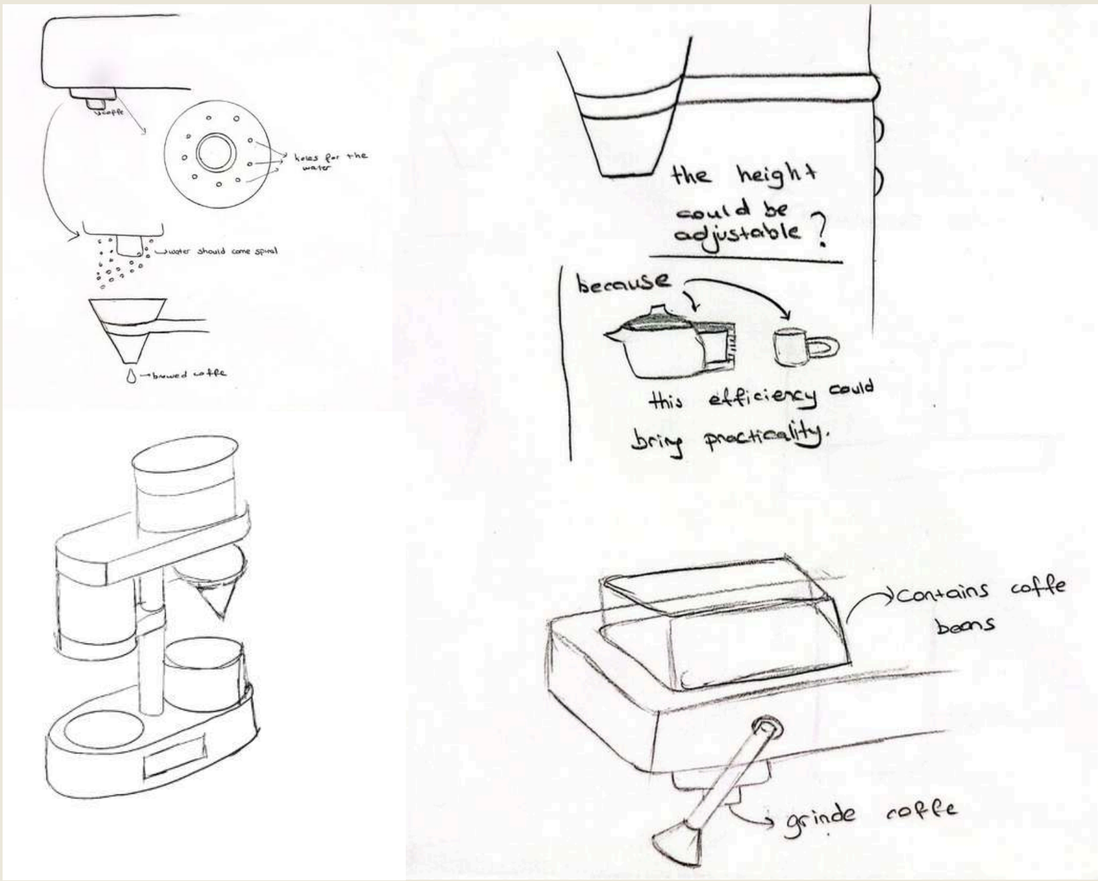
✕ Rejected

Complex or heavy forms that would disturb the minimal visual language.

🔄 Transformed

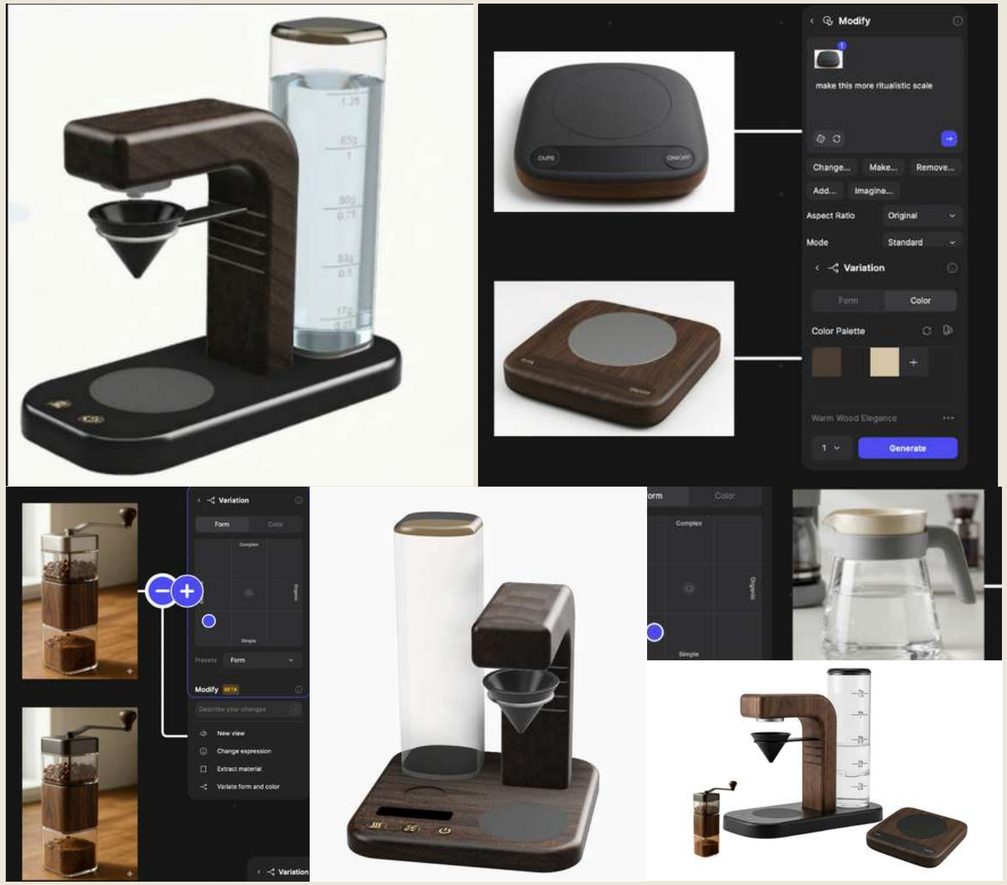
Refinement of spout and handle geometry to improve pouring ergonomics.

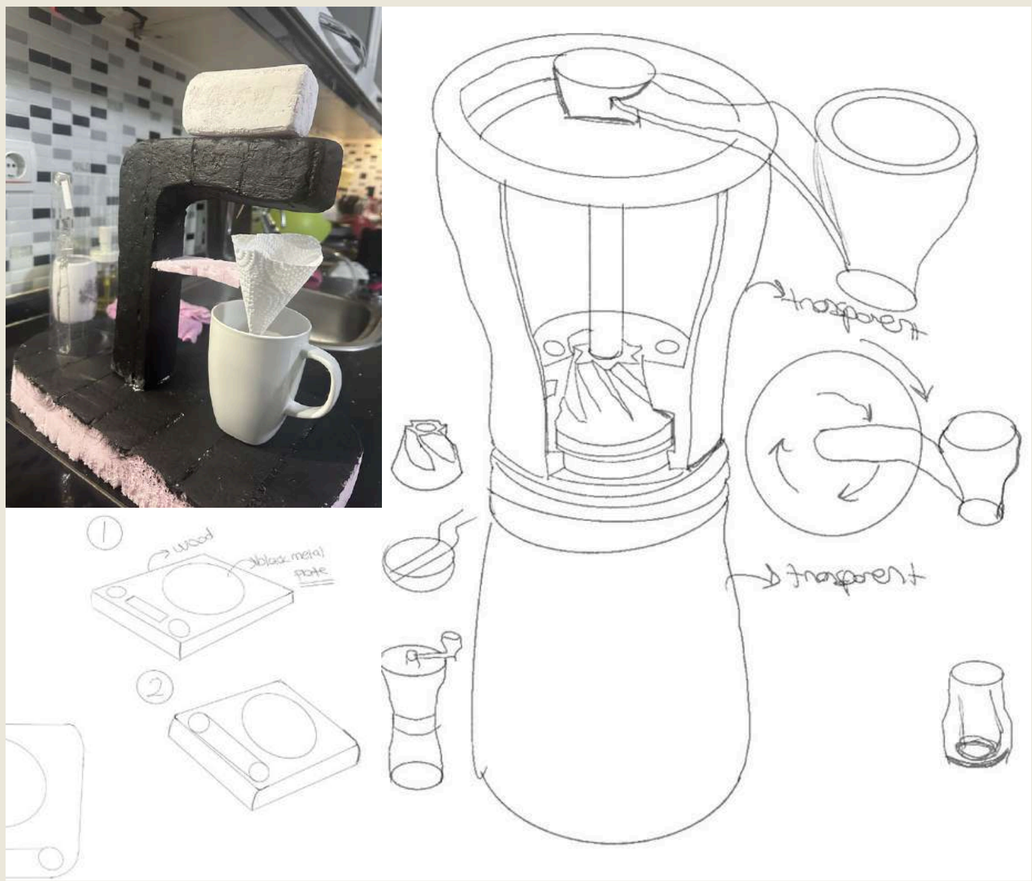




DESIGN DECISION
INITIAL

The design prioritizes a ritual-focused brewing experience where the transformation during brewing remains visible to the user. Fully automated systems that hide the extraction process were avoided to maintain user participation. Through iterative development, the early sketches were refined into a clearer pour-over station with improved workflow, proportions, and a minimal, calm form language.





DESIGN DECISION
REFINEMENT

The design evolved from a single brewing device into a complete pour-over coffee set. This decision was made to ensure that every stage of the preparation process—grinding, dosing, brewing, and serving—remains visible and actively experienced by the user. By structuring the system as a set rather a single system , the design strengthens the ritual aspect of coffee preparation and supports a more engaging and coherent brewing experience.

DESIGN DECISION
FINAL

The final design was developed as a ritual-focused pour-over coffee set that allows the user to experience every stage of the brewing process. Instead of hiding the preparation behind automation, the system keeps grinding, dosing, brewing, and serving visible and intentional. This approach strengthens the ritual aspect of coffee preparation and creates a coherent and engaging user experience.